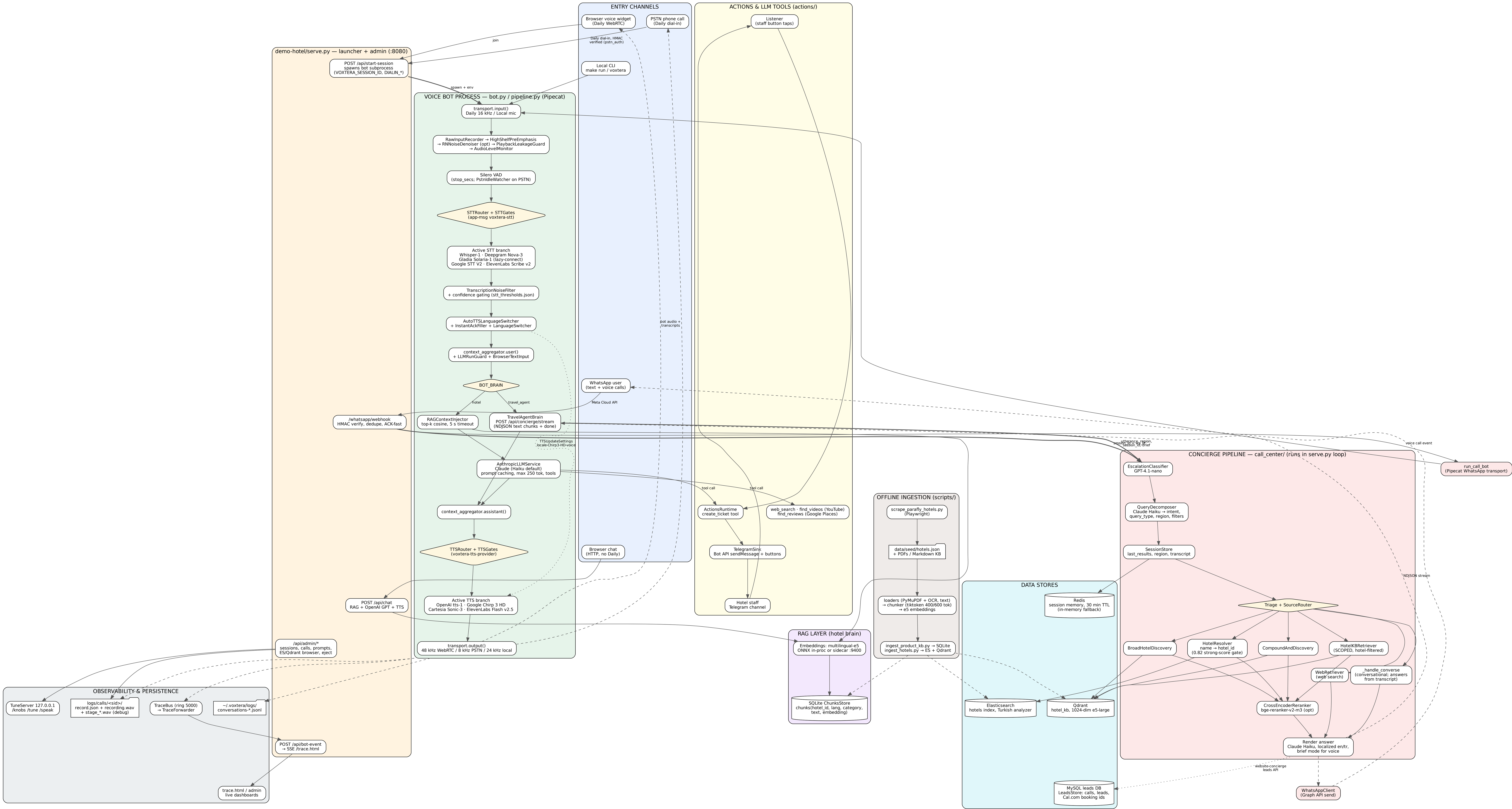
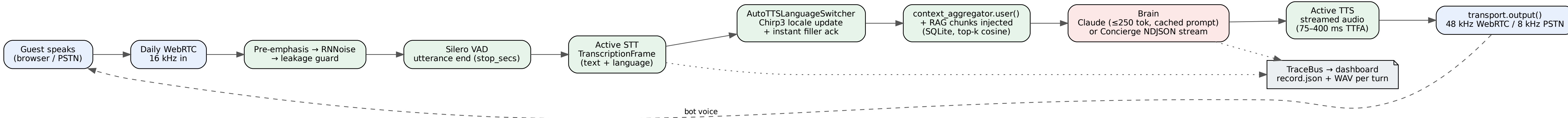


Voxtera — Full System Data Flow (2026-06-09)



Voxtera — One Voice Turn (latency path)



Key facts behind the arrows

Two brains (BOT_BRAIN). **hotel** = in-process RAG (SQLite + multilingual-e5) + Claude with tools. **travel_agent** = delegates each utterance to the ConciergePipeline over streaming NDJSON; emits the same frames as an LLM service so TTS and tracing are unchanged.

STT / TTS hot-swap. In Daily mode all providers run as parallel gated branches. Browser app-messages (voxtera-stt, voxtera-tts-provider) flip STTGate/TTSGate pairs. Gladia lazy-connects on activation to dodge the 1-session free-tier cap.

Language flow. STT auto-detects language per utterance; AutoTTSLanguageSwitcher rewrites the Chirp 3 HD voice id ({locale}-Chirp3-HD-{character}) mid-call; filler acks are language-matched.

WhatsApp. Text messages share the concierge with voice: wa_id is the session id, so Redis memory and region persist per contact. Voice-call events spawn a Pipecat WhatsApp call bot into the same pipeline.

Concierge retrieval paths. SCOPED (hotel KB, Qdrant), BROAD discovery, COMPOUND multi-constraint, RESOLVE (name to hotel_id via Elasticsearch, 0.82 strong-score gate), WEB search, and conversational (_handle_converse: answers from transcript, no retrieval).

LLM split. Escalation = GPT-4.1-nano. Decompose / render / converse = Claude Haiku. Voice bot = Claude (Haiku default) with prompt caching and a 250-token reply cap.

Persistence. Atomic record.json + stereo WAV per call (logs/calls/<sid>/), JSONL conversation audit (~/.voxtera/logs/), Redis sessions (30 min TTL, in-memory fallback), MySQL leads DB (website-concierge).

Observability. In-process TraceBus (ring 5000) forwards to serve.py /api/bot-event, broadcast over SSE to trace.html. Per-bot TuneServer (127.0.0.1) exposes /knobs, /tune, /speak for live edits.

Offline ingestion. scrape_parafly_hotels.py (Playwright) builds data/seed/hotels.json; loaders (PyMuPDF+OCR) feed the tiktoken chunker (400/600 tok) and e5 embeddings; ingest scripts populate SQLite, Elasticsearch, and Qdrant.